CVPR 2026 tutorial proposal

Monte Carlo simulation of partial differential equations

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(Bios and other background for the proposers are included in the last page)

Primary subject area: Vision + graphics

Secondary subject areas: Physics-based vision and shape-from-X, Embodied vision: simulation

Preference: half-day event **Presentation format:** in-person **Special requests:** none

Tutorial description

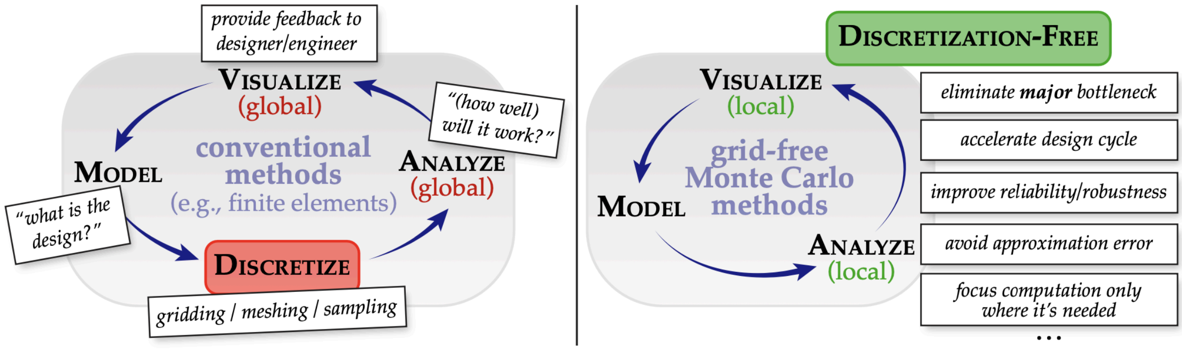

Figure 1: Grid-free Monte Carlo methods for PDE simulation can significantly speedup the engineering design cycle by eliminating the major bottleneck of discretization in conventional solvers for partial differential equations (left). Lifting the dependence on discretization and global solves improves robustness, and like Monte Carlo ray tracing allows for computational effort to be focused entirely on local regions of interest (right).

This tutorial provides a detailed overview of grid-free Monte Carlo methods for solving partial differential equations (PDEs) based on the walk on spheres (WoS) algorithm, with emphasis on problems with high geometric complexity and applications of physics simulation in computer vision. PDEs are a basic building block of models and algorithms used throughout science, engineering, and visual computing, and have recently been finding widespread adoption within vision pipelines for inverse problems and robotics applications. Yet despite decades of research, conventional PDE solvers struggle to capture the immense geometric complexity of the natural world. A perennial challenge is spatial discretization: traditionally, one must partition the domain into a high-quality volumetric mesh—a process that can be brittle, memory intensive, and difficult to parallelize. WoS makes a radical departure from this approach, by reformulating the problem in terms of recursive integral equations that can be Monte-Carlo estimated, eliminating the need for spatial discretization. As these equations strongly resemble those in light transport, one can leverage knowledge from ray tracing and Monte Carlo rendering to develop new PDE solvers that share many of their advantages: no meshing, trivial parallelism, end-to-end differentiability, and evaluation at any point without global solves.

The tutorial will cover fundamentals of Monte Carlo algorithms such as walk on spheres and walk on stars that solve PDEs such as the Poisson equation. Topics include formulating the solution as an integral equation, generating samples via recursive random walks, and employing accelerated distance and ray intersection queries to efficiently handle complex geometries. The tutorial will then cover advanced topics including variance reduction, differentiable and multi-physics simulation, and applications in industry, vision, and graphics. Participants will gain experience setting up demo applications involving data interpolation, heat transfer, and geometric optimization using the open-source "Zombie" library. Lastly, the tutorial will include a hands-on tutorial on implementing walk on spheres and walk on stars from scratch in a few minutes.

Tutorial outline

- **Introduction and tutorial overview** (15 minutes)
    - Broad applicability of PDEs and challenges with conventional grid-based solvers
    - Photorealistic rendering: From finite element radiosity to Monte Carlo ray tracing
    - Tutorial overview
- **Basics of Monte Carlo integration** (15 minutes)
    - Integral estimation: quadrature and aliasing
    - Monte Carlo estimation: bias, consistency, variance, sample generation
    - Numerical and computational considerations
- **Walk on spheres for solving elliptic equations** (30 minutes)
    - Recursive integral formulation
    - Closest point queries
    - Extension to PDEs with variable material coefficients and participating media
- **Walk on stars for Neumann and Robin boundary conditions** (30 minutes)
    - Boundary integral equation
    - Closest silhouette point queries
    - Extension to Robin boundary conditions
- **Hands-on tutorial: implementing walk on stars from scratch** (30 minutes)
- **Demos and evaluation** (30 minutes)
    - Stress tests on complex geometric domains
    - Comparisons to grid-based PDE solvers
    - Coupled ray tracing and walk on spheres
- **Variance reduction** (30 minutes)
    - Importance sampling and bidirectional schemes
    - Boundary value caching, harmonic caching, mean value caching
    - Neural caching and control variates
- **Differentiability** (30 minutes)
    - Differential walk on spheres for shape optimization
    - Differential walk on spheres for material estimation
    - Differential walk on stars for Robin boundary conditions
- **Applications** (15 minutes)
    - Monte Carlo fluid simulation
    - Coupled Monte Carlo heat transfer simulation for vision with light and heat
    - Walk on spheres for robotics and neural PDE solvers
- **Q&A session** (15 minutes)
    - Current limitations, future directions, and opportunities for vision and robotics

Resources to be distributed to attendees

The tutorial will provide resources including: presentation slides (example), video recording (example), literature (example), implementations (example), programming tutorials (example).

Educational value, interest, relevance, and diversity

Physics simulation has become an integral part of modern computer vision and robot learning systems, with a proliferation of pipelines that combine partial differential equation (PDE) solvers, learning-based modules, and vision components in an end-to-end optimizable fashion. Unfortunately, the use of traditional grid-based solvers (e.g., finite elements, boundary elements, finite differences) in such pipelines creates memory and compute bottlenecks that severely constrain scalability to massively complex geometry or massively parallel compute. Monte Carlo PDE solvers promise to overcome these bottlenecks, featuring logarithmic scaling with geometric complexity, embarrassing parallelization, and end-to-end differentiability. These new solvers have been rapidly developing in the past five years in graphics, but remain relatively unknown within vision.

This tutorial will help bridge this gap between graphics and vision, by presenting a comprehensive overview of Monte Carlo physics simulation. To achieve its knowledge transfer goals, the tutorial will cover theory, algorithms, implementations, and applications, and highlight opportunities for vision and robotics applications. The tutorial will additionally feature programming resources and hands-on tutorials for vision researchers and engineers to get started using Monte Carlo PDE solvers. To ensure that the tutorial will be inclusive of all perspectives in physics simulation, it will highlight works from a large diversity of researchers beyond the presenters, from both academia and industry, that have been actively developing methods and applications (e.g., "Variance reduction," "Differentiability," and "Applications" parts of the tutorial outline).

Overall, this tutorial will serve as a new point of convergence between graphics and vision, and facilitate cross-pollination of ideas between researchers in these areas. The widespread adoption of Monte Carlo PDE solvers within computer vision has the potential for transformative impact similar to that enabled by the widespread adoption of Monte Carlo rendering within vision pipelines.

Expected target audience

The intended audience includes researchers and practitioners in computer vision, computer graphics, computational photography, and robotics, who are interested in using scalable and differentiable physics simulation to solve inverse problems, or integrate physics in machine learning, reinforcement learning, robotics, and other inferential pipelines—especially in settings involving geometry complexity that is intractable with current grid-based physics solvers. The estimated number of attendees is medium (100–300 attendees).

Relationship to previous tutorials and short courses

The proposed topic has not been presented in previous tutorials at CVPR, ICCV, or ECCV. It has been presented in courses at SGP 2024, SIGGRAPH 2025, and SIGGRAPH Asia 2025 (see next).

Recordings of previous talks by the presenters

Available recordings from previous courses and tutorials offered by the proposers: CVPR 2021, CVPR 2023, SIGGRAPH 2023, SGP 2024, SIGGRAPH 2025.

References

[1] Sawhney and Crane, "Monte Carlo Geometry Processing: A Grid-Free Approach to PDE-Based Methods on Volumetric Domains," ACM Transactions on Graphics 2020.

[2] Sawhney*, Seyb*, Jarosz†, Crane†, "Grid-Free Monte Carlo for PDEs with Spatially Varying Coefficients," ACM Transactions on Graphics 2022.

[3] Sawhney*, Miller*, Gkioulekas†, Crane†, "Walk on Stars: A Grid-Free Monte Carlo Method Method for PDEs with Neumann Boundary Conditions," ACM Transactions on Graphics 2023.

[4] Sawhney and Miller, "Zombie: A Monte Carlo PDE solver," https://github.com/rohan-sawhney/zombie.

[5] Miller*, Sawhney*, Crane†, Gkioulekas†, "Boundary Value Caching for Walk on Spheres," ACM Transactions on Graphics 2023.

[6] Miller*, Sawhney*, Crane†, Gkioulekas†, "Walkin' Robin: Walk on Stars with Robin Boundary Conditions," ACM Transactions on Graphics 2024.

[7] Miller, Sawhney, Crane, Gkioulekas, "Differential Walk on Spheres," ACM Transactions on Graphics 2024.

[8] Miller, Sawhney, Crane, Gkioulekas, "Solving partial differential equations in participating media," ACM Transactions on Graphics 2025.

[9] Yu, Sawhney, Miller, Wu, Zhao, "Robust Derivative Estimation with Walk on Stars," ACM Transactions on Graphics 2025.

[10] Zhou, d'Eon, Sawhney, Jarosz, "Harmonic caching for walk on spheres," ACM Transactions on Graphics 2025.

Proposers' bios and experience

Rohan Sawhney (http://rohansawhney.io/) is a Senior Research Scientist in NVIDIA's High Fidelity Physics group. He earned his PhD in Computer Science from CMU under Keenan Crane. His research focuses on developing efficient and reliable Monte Carlo algorithms inspired by photorealistic rendering for PDE-based geometric computing, eliminating the need for volumetric mesh generation. Rohan has received Dissertation and Best Paper Awards at SIGGRAPH and CMU for his work on Monte Carlo PDE solvers, as well as a Symposium on Geometry Processing Best Software Award for Boundary First Flattening, an application to quickly and robustly generate UV maps.

Bailey Miller (https://www.bailey-miller.com/) is a PhD student at CMUadvised by Ioannis Gkioulekas. His research develops Monte Carlo algorithms for large-scale simulation and stochastic scene representations for inverse tasks like 3D reconstruction. His work has received recognition through the NSF Graduate Research Fellowship, the NVIDIA Graduate Research Fellowship, and best paper awards at both SIGGRAPH and CVPR. He has previously co-taught course on Monte Carlo PDE solvers at the Symposium on Geometry Processing, SIGGRAPH, and SIGGRAPH Asia, and has collaborated with industry through internships at Adobe, Apple's XDG (Exploratory Design Group), and NVIDIA's High Fidelity Physics team.

Ioannis Gkioulekas (http://www.cs.cmu.edu/~igkioule) is an associate professor at Carnegie Mellon University (CMU), where he works broadly in computer graphics, focusing on computational imaging, physically based rendering, and physical simulation. Recent topics of interest within these areas include: non-line-of-sight imaging, single-photon imaging, LIDAR, SONAR, interferometry, acousto-optics, differentiable rendering, Monte Carlo simulation, and stochastic geometry. He has previously taught courses and tutorials on these topics, both at CMU and at venues including CVPR 2021, CVPR 2023, SIGGRAPH 2023, and SIGGRAPH 2025. For his research, he has received the NSF CAREER Award, the Sloan Research Fellowship, the Bodossaki Distinguished Young Researcher Award, and best paper awards at CVPR and SIGGRAPH.

Keenan Crane (https://www.cs.cmu.edu/~kmcrane/) is the Michael B. Donohue Associate Professor of Computer Science and Robotics at Carnegie Mellon University, where he leads the Geometry Collective and is a member of the Center for Nonlinear Analysis. His work uses insights from differential geometry to build fundamental representations and algorithms for processing, designing, and analyzing geometric data. Keenan received a BS from UIUC, was a Google PhD Fellow at Caltech, an NSF Mathematical Sciences Postdoctoral Fellow at Columbia University. He is a Packard Fellow, recipient of an NSF CAREER Award and Best Paper Awards at SIGGRAPH.

The proposers have been collaborating over the past five years on developing grid-free Monte Carlo methods for simulation of partial differential equations as a new research direction within computer graphics and computer vision. They have coauthored nine papers on research topics within this area—two of which have been recognized with best paper awards—developing new simulation capabilities, including support for different boundary conditions, differentiability, simulation in volumetric media, and acceleration schemes using caching or improved importance sampling. To encourage dissemination and widespread adoption of Monte Carlo simulation methods, they have released open-source software implementations for Monte Carlo simulation methods, and created instructional materials presented as tutorials and courses at several venues (Symposium on Geometry Processing 2024, SIGGRAPH 2025, SIGGRAPH Asia 2025). These dissemination efforts have resulted in collaboration with industry partners looking to adopt the new simulation methods. Representative publications and other resources are included in the references.